

O-RAN Near-Real-Time RIC (xApp Container)

**5G NR gNB
& UE Mobility**

E2SM-KPM

RSRP Telemetry
 $\mathbf{R} \in \mathbb{R}^{50}$

**Pillar 1: Kinematic
Physics Layer**

$$v_t = \frac{dr}{dt}$$
$$a_t = \frac{d^2r}{dt^2}$$

Energies (E_v, E_a)
 $\rightarrow \mathbf{X}_{\text{kin}} \in \mathbb{R}^{64}$

**Pillar 3: TinyML
Quantized Engine**

INT8 MLP (56.8% RAM ↓)
Pruned DT (0.096 ms)
 $\rightarrow \text{Prob. } \hat{p} \in [0, 1]$

**Pillar 2: DAT
Cost Engine**

$C(P_{\text{th}}) = \text{FP} + 5\text{FN}$
Opt. $P_{\text{th}}^* = 0.35$
81.9% RLF Drop ↓
 $\rightarrow \text{HO Decision}$